Pizza Sales Data Analysis

1. Project Overview

This project analyzes pizza sales data using transactional data from 21350 order in a Pizza Hut store. The goal is to uncover insights into pizza sales, peak hours in a day, average number of orders per day, top ordered pizza types, each pizza category contribution in revenue to guide strategic business decisions.

2. Dataset Summary

- 4 Datasets pizzas, pizza_types, orders, order_details
- 21350 Row orders
- 48620 Row order_details
- Key Features:
    - Pizzas details (pizza_id, pizz_type_id, size, price)
    - Pizza type details (pizza name, category, ingredients)
    - Orders detail (order_id, date, time)
    - Order details data (order_detail_id, quantity)

3. Data Analysis using SQL (Business Transactions)

I performed structured analysis in MySQL to answer key business questions:

1. **Total Orders placed** – Found total number of orders placed

	total_orders
▶	21350

2. **Total Revenue from Pizza Sales** – Calculated total revenue from pizza sales.

	revenue
▶	817860.05

3. **Highest- priced Pizza** – Identified the highest-priced pizza.

	name	price
▶	The Greek Pizza	35.95

4. **Most common pizza size –** Identified the most common pizza size ordered.

	size	order_quantity
▶	L	18526

5. **Top 5 ordered Pizza types –** Found the top 5 most ordered pizza types along with their quantities.

	name	order_quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

6. **Category-wise Pizzas** -Found the category-wise distribution of pizzas.

	category	COUNT(name)
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

7. **Category wise Total quantity of each pizza ordered** - Calculated the total quantity of each pizza category ordered.

	category	order_quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

8. **Orders by day hours** – Determined the distribution of orders by hour of the day.

hours	orders
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8
9	1

9. **Average Pizzas ordered per day** – Grouped the orders by date and calculated the average number of pizzas ordered per day.

age_pizzas_ordered_per_day
138

10. **Top 3 ordered Pizza types based on revenue** – Listed the top 3 most ordered pizza types based on revenue.

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

11. **Each pizza type percentage contribution to revenue** – Calculated the percentage contribution of each pizza type to total revenue.

category	revenue
Veggie	23.68
Chicken	23.96
Supreme	25.46
Classic	26.91

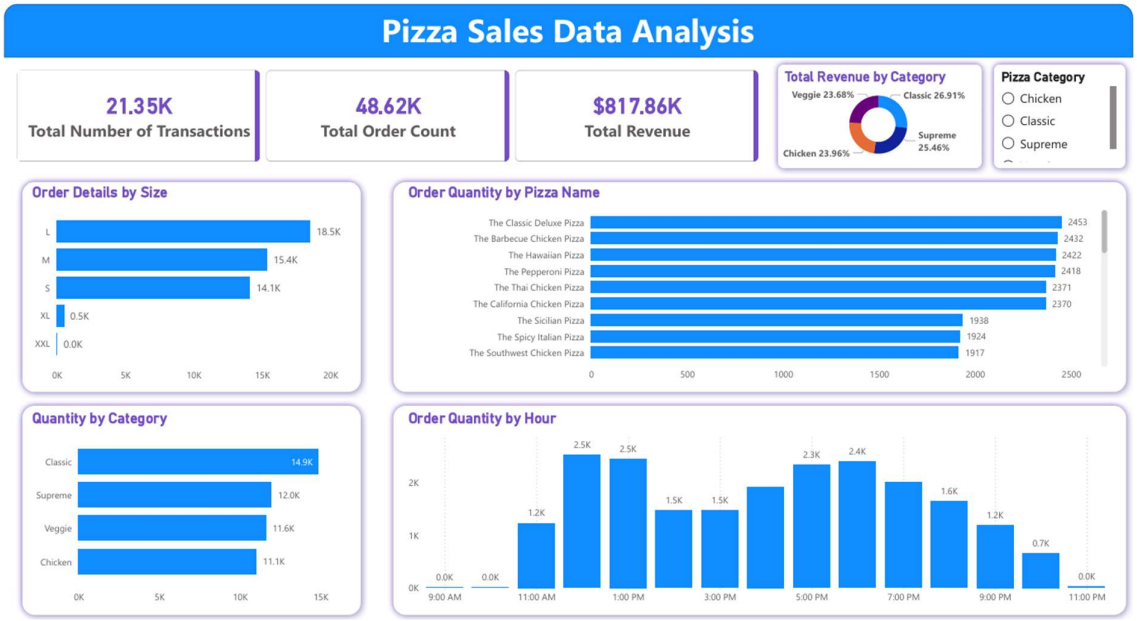
12. **Cumulative revenue** – Analyzed the cumulative revenue generated over time.

order_date	cum_revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.350000000002
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.300000000003
2015-01-14	32358.700000000004
2015-01-15	34343.50000000001

358 row(s) returned

13. **Each Category Top 3 ordered Pizza types based on revenue** – Listed the top 3 most ordered pizza types based on revenue for each pizza category.

category	name	ranks
Chicken	The Thai Chicken Pizza	1
Chicken	The Barbecue Chicken Pizza	2
Chicken	The California Chicken Pizza	3
Classic	The Classic Deluxe Pizza	1
Classic	The Hawaiian Pizza	2
Classic	The Pepperoni Pizza	3
Supreme	The Spicy Italian Pizza	1
Supreme	The Italian Supreme Pizza	2
Supreme	The Sicilian Pizza	3
Veggie	The Four Cheese Pizza	1
Veggie	The Mexicana Pizza	2
Veggie	The Five Cheese Pizza	3

4. Dashboard in Power BI

I've built interaction dashboard in Power BI to present insights visually.